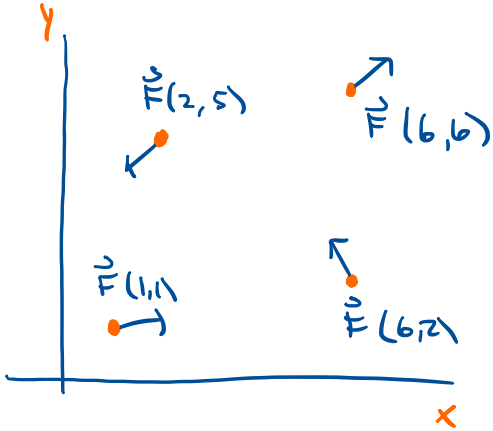
17.3 Vector fields

A **vector field** in $\mathbb{R}^2$ is a function

$$\vec{F}(x,y) = \langle F_1(x,y), F_2(x,y) \rangle$$

It assigns vectors to each point in $\mathbb{R}^2$. Vector fields in $\mathbb{R}^3$ are defined similarly, with a third component function $F_3(x,y,z)$.



→ Do problems 1–4.

Recall that if $z = f(x,y)$ is a function, then its gradient is

$$\nabla f(x,y) = \langle f_x(x,y), f_y(x,y) \rangle.$$

This is a vector field! Any vector field which is the gradient of a function is called a **gradient field**.

→ Do problem 5.